

EUCLID ASCENT

ALGEBRA AND FUNCTIONS

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1. Which of the following is equal to

$$\frac{1}{\sqrt{5}-2}?$$

- A. $\sqrt{5} + 2$
- B. $2 - \sqrt{5}$
- C. $\frac{\sqrt{5} + 2}{2}$
- D. $\sqrt{5} - 2$
- E. $5 + 2\sqrt{5}$
- F. 9

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2. The roots of

$$x^2 + 4x - 1 = 0$$

are p and q . What is $(p + 1)(q + 1)$?

- A. -6
- B. -2
- C. -4
- D. 2
- E. 0
- F. 4

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3. How many integers x with $0 \leq x \leq 6$ satisfy

$$(x - 1)(x - 4) < 0?$$

- A. 1
- B. 3
- C. 4
- D. 5
- E. 6
- F. 2

1. How many real solutions does the equation

$$|x - 2| = (x - 2)^2$$

have?

- A. 5
- B. 0
- C. 1
- D. 2
- E. 4
- F. 3

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2. A polynomial P satisfies $P(2) = 5$. What is the remainder when

$$(x + 1)P(x)$$

is divided by $x - 2$?

- A. 5
- B. 12
- C. 7
- D. 10
- E. 15
- F. 25

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3. For which real values of x is

$$\frac{x^2 - 1}{x - 1} = x + 1?$$

- A. All real x
- B. All real x except 1
- C. Only $x = 1$
- D. Only $x = -1$
- E. All real x except -1
- F. No real x

1. A function P is defined for every real number and satisfies

$$P(x+1) - P(x) = 2x + 1$$

for every real x . Given that $P(0) = 3$, what is $P(2)$?

- A. 5
- B. 6
- C. 7
- D. 8
- E. 9
- F. 11

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2. The function f is defined by

$$f(x) = x^2 - 4x + 7.$$

A real number $a \neq 6$ satisfies $f(a) = f(6)$. What is a ?

- A. -6
- B. -4
- C. 0
- D. -2
- E. 2
- F. 4

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3. A real number x satisfies

$$2^x + 2^{-x} = 3.$$

What is

$$4^x + 4^{-x}?$$

- A. 9
- B. 7
- C. 5
- D. 11
- E. 6
- F. 8

1. A quadratic polynomial P satisfies

$$P(0) = 8, \quad P(1) = P(3) = 5.$$

What is $P(2)$?

- A. 2
- B. 3
- C. 5
- D. 6
- E. 4
- F. 7

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2. A polynomial $P(x)$ leaves remainder 2 when divided by $x - 1$, and remainder -2 when divided by $x + 1$. What is the remainder when $P(x)$ is divided by $x^2 - 1$?

- A. $2x$
- B. 2
- C. -2
- D. $-2x$
- E. $2x + 2$
- F. $2x - 2$

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3. What is the greatest possible value of

$$\frac{2x + 4}{x^2 + 4x + 8}, \quad x \in \mathbb{R}?$$

- A. $-\frac{1}{2}$
- B. 0
- C. $\frac{1}{4}$
- D. $\frac{1}{2}$
- E. 1
- F. 2

1. A polynomial P with real coefficients satisfies

$$[P(x)]^2 = P(x)$$

for every real number x .

Which statement must be true?

- A. $P(x) = 0$ for every x .
- B. $P(x) = 1$ for every x .
- C. Every constant polynomial works.
- D. P is identically 0 or identically 1.
- E. There are infinitely many non-constant possibilities.
- F. $P(x) = x$ is also possible.

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2. A positive real number x satisfies

$$x = 2 + \frac{1}{2 + \frac{1}{x}}.$$

What is x ?

- A. 3
- B. $\sqrt{2} - 1$
- C. $1 + \sqrt{2}$
- D. $\sqrt{2}$
- E. $2 + \sqrt{2}$
- F. 1

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3. Positive real numbers x and y satisfy

$$x + y = xy$$

and

$$x^2 + y^2 = 10.$$

What is xy ?

- A. $1 - \sqrt{11}$
- B. $1 + \sqrt{11}$
- C. $2 + \sqrt{11}$
- D. 5
- E. 10
- F. $\sqrt{11}$

1. The equation

$$|x^2 - 4| = |x|$$

has two positive solutions p and q .
What is $p + q$?

- A. 4
- B. 3
- C. $\sqrt{17}$
- D. $\sqrt{13}$
- E. $\sqrt{15}$
- F. $2\sqrt{5}$

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2. Let

$$f(x) = \frac{x^2 - 4x + 3}{x^2 - 4x + 5}, \quad x \in \mathbb{R}.$$

Which of the following describes exactly the possible values of $f(x)$?

- A. $(-1, 1)$
- B. $(-\infty, 1)$
- C. $(-1, 1]$
- D. $[-1, 1]$
- E. $[-1, 1)$
- F. $[-1, \infty)$

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3. A quadratic polynomial P with real coefficients satisfies

$$P(P(x)) = x^4 + 2x^2 + 2$$

for every real x .
What is $P(1)$?

- A. 2
- B. 0
- C. 1
- D. 3
- E. 4
- F. It cannot be determined

1. Let

$$f(x) = (x - 2)^2 + 1.$$

How many real solutions does

$$f(f(x)) = f(x)$$

have?

- A. 4
- B. 1
- C. 0
- D. 2
- E. 3
- F. 6

2. For which real values of k is

$$x^4 - kx^3 + kx - 1 > 0$$

true for every real $x > 1$?

- A. $k < -2$
- B. $k \leq 2$
- C. $k \leq -2$
- D. $-2 \leq k \leq 2$
- E. $k < 2$
- F. $k \geq 2$

3. Real constants a, b, c define

$$f(x) = \frac{ax + b}{cx - a}$$

whenever the denominator is non-zero.

It is known that $f(3) = 7$, and that $f(7)$ is defined. What is $f(7)$?

- A. $\frac{1}{3}$
- B. 1
- C. 7
- D. 3
- E. 10
- F. It cannot be determined

1. For real numbers p and q , the polynomial

$$P(x) = (x^2 + px + 1)(x^2 + qx + 1)$$

has four distinct positive real roots.

When $P(x)$ is written as

$$x^4 + ax^3 + bx^2 + ax + 1,$$

Which is the strongest conclusion that must hold about b ?

- A. $b > 0$
 - B. $b \geq 2$
 - C. $b > 4$
 - D. $b \geq 6$
 - E. $b < -6$
 - F. $b > 6$
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2. What is the greatest possible value of

$$\frac{x^2 + 6x + 1}{x^2 + 2x + 5}, \quad x \in \mathbb{R}?$$

- A. 1
 - B. $\frac{3}{2}$
 - C. 2
 - D. $1 + \sqrt{2}$
 - E. $\sqrt{2}$
 - F. There is no greatest value
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3. Non-zero real numbers u, v, w satisfy

$$(x - u)(x - v)(x - w) = x^3 - ux^2 + vx - w$$

for every real x .

What is $u + v + w$?

- A. -3
- B. -1
- C. 0
- D. 1
- E. 2
- F. 3

1. For which real value of k does the equation

$$x^4 - 8x^3 + (16 - k)x^2 + 4kx + 4k = 0$$

have exactly three distinct real solutions?

- A. -8
- B. -4
- C. -2
- D. 0
- E. 2
- F. 4

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2. A quadratic polynomial P with leading coefficient 1 has the property that its two roots are

$$P(1) \quad \text{and} \quad P(-1).$$

What is the product of all possible values of $P(0)$?

- A. -3
- B. -1
- C. $-\frac{1}{3}$
- D. $\frac{1}{3}$
- E. 1
- F. 3

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3. A polynomial P has degree at most 5 and satisfies

$$P(k) = \frac{1}{k+1}$$

for $k = 0, 1, 2, 3, 4, 5$.

What is $P(6)$?

- A. $-\frac{1}{7}$
- B. $\frac{1}{7}$
- C. $\frac{1}{6}$
- D. 1
- E. It cannot be determined
- F. 0